Question			Answer	Marks	AO	Guidance
10			$4^2 + 1^2 + x^2$	**B1**	**1.1**	• Use of Pythagoras implied by $17 + x^2$
			$17 + x^2 = n^2$ OR $n^2 - x^2 = 17$	**B1**	**1.1**	• Understanding that $17 + x^2$ is a square number soi e.g. by equating to at least one square number $17 + x^2 = 25$ • Allow e.g. $n = \sqrt{17 + x^2}$ and n integer soi e.g. by equating to an integer
			$(n - x)(n + x) = 17$	**M1**	**3.1a**	• Rearranging to get *their* 17 on one side and factorising
			$n - x = 1$ **and** $n + x = 17$ AND $n - x = -1$ **and** $n + x = -17$	**A1**	**2.2a**	• OR $n - x = 17$ **and** $n + x = 1$ AND $n - x = -17$ **and** $n + x = -1$ • Implies **M1.** If the final **B2** is earned following only one pair of correct equations then award **A1**
			Alternative method 1 for M1 A1 Evidence of adding square numbers to 17 to try to get a square number Or subtracting 17 from square numbers to get a square number Or $x^2 = 64$	**M1**		• At least two different square numbers added to *their* 17 e.g. two from: $17 + 1$, $17 + 4$, $17 + 9$, $17 + 16$ $17 + 25$, $17 + 49$, $17 + 64$, $17 + 81$ and so on • Or tested two using $17 + x^2 = 25$, $17 + x^2 = 36$, $17 + x^2 = 49$, $17 + x^2 = 64$, $17 + x^2 = 81$ and so on
			Explanation of why there will be no further values	**A1**		• e.g. Since 64 and 17 + 64 are adjacent square numbers there will be no larger values of x^2
			Alternative method 2 for M1 A1 $(x + 1)^2 - x^2 = 17$	**M1**		• Uses adjacent square numbers • Implied by $2x + 1 = 17$ • Condone not testing earlier squares for this **M1**
			$2x + 1 = 17$ and $(x - 1)^2 - x^2 = 17$	**A1**		• If the final **B2** is earned following **M1** then award **A1**
			8 and -8	**B2** **[6]**	**1.1**	• Both values and no others • **B1** if only one value given (ignore extras)